

8.1 Consider the op-amp circuits shown in Fig. 1, where the op-amp has infinite input resistance and zero output resistance but finite open-loop gain A_0 .

- Write down the expressions for feedback factor β , loop gain, and closed-loop gain A_{CL} for the amplifiers in Fig. 1(a) and Fig. 1(b), respectively.
- For the 2 amplifiers, if $R_1 = 10\text{K}\Omega$, and we want $|A_{CL}| = 10$, find the nominal value (assuming $A_0 = \infty$) and the exact value of R_2 for 2 cases: (i) $A_0 = 1000$; (ii) $A_0 = 200$.
- For each of the 2 cases in (b), find the percentage change in $|A_{CL}|$ that results when A decreases by 20%. Comment on the results.

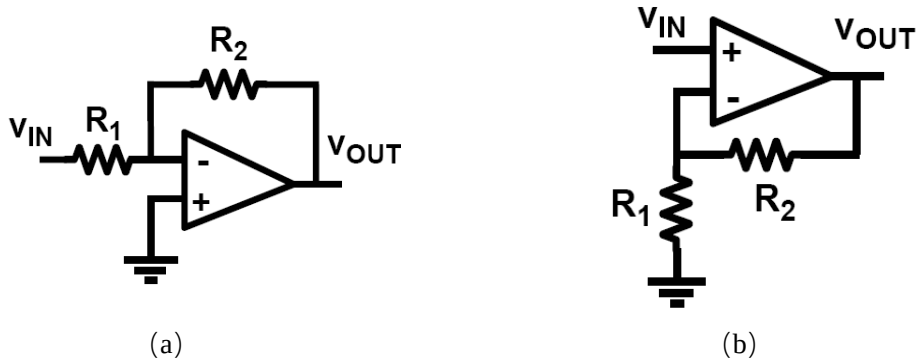


Fig. 1.

(a)

Fig1 (a):

$$A_{open} = \frac{R_2}{R_1 + R_2} A_0$$

$$\text{loop gain} = \frac{R_1}{R_1 + R_2} A_0$$

$$A_{CL} = \frac{A_{open}}{1 + \text{loop gain}} = \frac{R_2}{R_1} \frac{\frac{R_1}{R_1 + R_2} A_0}{1 + \frac{R_1}{R_1 + R_2} A_0}$$

$$\beta = \frac{R_1}{R_2}$$

Fig1 (b):

$$A_{open} = A_0$$

$$\text{loop gain} = \frac{R_1}{R_1 + R_2} A_0$$

$$A_{CL} = \frac{A_{open}}{1 + \text{loop gain}} = \frac{1}{\frac{R_1}{R_1 + R_2}} \frac{\frac{R_1}{R_1 + R_2} A_0}{1 + \frac{R_1}{R_1 + R_2} A_0}$$

$$\beta = \frac{R_1}{R_1 + R_2}$$

(1.1)

(b)

Fig1(a):

$$A_{CL} = \frac{A_{open}}{1 + loop\ gain} = \frac{R_2}{R_1} \frac{\frac{R_1}{R_1 + R_2} A_0}{1 + \frac{R_1}{R_1 + R_2} A_0} = 10$$

$$A_0 = \infty \quad R_2 = 100k$$

$$A_0 = 1000 \quad R_2 = 101.11k$$

$$A_0 = 200 \quad R_2 = 105.79k$$

Fig1(b):

$$A_{CL} = \frac{A_{open}}{1 + loop\ gain} = \frac{1}{\frac{R_1}{R_1 + R_2}} \frac{\frac{R_1}{R_1 + R_2} A_0}{1 + \frac{R_1}{R_1 + R_2} A_0} = 10$$

$$A_0 = \infty \quad R_2 = 90k$$

$$A_0 = 1000 \quad R_2 = 91.01k \quad \frac{R_1}{R_1 + R_2} = 0.099$$

$$A_0 = 200 \quad R_2 = 95.26k \quad \frac{R_1}{R_1 + R_2} = 0.095 \quad (1.2)$$

(c)

Fig1(b):

$$A_0 = 800 \quad \frac{R_1}{R_1 + R_2} = 0.099 \quad A_{CL} = \frac{A_{open}}{1 + loop\ gain} = \frac{1}{\frac{R_1}{R_1 + R_2}} \frac{\frac{R_1}{R_1 + R_2} A_0}{1 + \frac{R_1}{R_1 + R_2} A_0} = 9.975$$

$$percentage = \frac{10 - A_{CL}}{10} = 2.5\% \quad (1.3)$$

$$A_0 = 160 \quad \frac{R_1}{R_1 + R_2} = 0.095 \quad A_{CL} = \frac{A_{open}}{1 + loop\ gain} = \frac{1}{\frac{R_1}{R_1 + R_2}} \frac{\frac{R_1}{R_1 + R_2} A_0}{1 + \frac{R_1}{R_1 + R_2} A_0} = 9.501$$

$$percentage = \frac{10 - A_{CL}}{10} = 4.99\%$$

8.2 It is required to design a amplifier with an **accurate low-frequency gain** of 1000 and a 3-dB frequency of 1 MHz. You have available gain stages with a gain of 1000 but with a dominant high-frequency pole at 20 kHz. Provide a design that employs a number (N) of such stages in cascade, each with negative feedback of an appropriate feedback factor (β). Please determine the minimum N and exact value of β .

Hint: The 3-dB frequency of a cascade of N identical gain stages, each with a 3-dB frequency $f_{3dB|stage}$ is given by

$$f_{-3dB|cascade} = \sqrt{2^{1/N} - 1} \cdot f_{-3dB|stage}$$

$$f_{-3dB|stage} = (1 + \beta A_0) w_0 \quad A_0 = 1000 \quad w_0 = 20k$$

$$f_{-3dB|cascade} = \sqrt{2^{1/N} - 1} \cdot f_{-3dB|stage} \quad f_{-3dB|cascade} = 1M = 1000k$$

$$A_{stage} = \frac{A_0}{1 + A_0 \beta}$$

$$A_{cascade} = (A_{stage})^N \quad A_{cascade} = 1000$$

$$N = 2$$

$$A_{stage} = \sqrt{1000} = 31.6228 \Rightarrow 1 + A_0 \beta = 31.6228$$

$$f_{-3dB|stage} = (1 + \beta A_0) w_0 = 31.6228 \times 20 \times 1000 = 632456$$

$$f_{-3dB|cascade} = \sqrt{2^{1/N} - 1} \cdot f_{-3dB|stage} = 407045$$

$$N = 3$$

$$A_{stage} = (1000)^{\frac{1}{3}} = 10 \Rightarrow 1 + A_0 \beta = 100$$

$$f_{-3dB|stage} = (1 + \beta A_0) w_0 = 100 \times 20 \times 1000 = 2 \times 10^6$$

$$f_{-3dB|cascade} = \sqrt{2^{1/N} - 1} \cdot f_{-3dB|stage} = 1.02 \text{ MHz}$$

$$\beta = 0.099$$

(2.1)

8.3 In Fig. 3, assume that $(W/L)_1 = 50/0.5$, $(W/L)_2 = 100/0.5$, $R_D = 2 \text{ k}\Omega$, $C_2 = 3C_1$, $\mu_n C_{OX} = 50 \times 10^{-6} \text{ A/V}^2$, $V_{TH} = 0.7 \text{ V}$. Neglecting channel-length modulation and body effect, answer the following questions:

- Determine the feedback type. (Choose one from VVF, CVF, VCF and CCF)
- Derive the expression of loop gain.
- Calculate the bias current of M1 and M2 such that the input resistance at low frequencies is equal to $50 \text{ }\Omega$.

Hint: you may need to study a few materials in the textbook or the lecture notes that are not taught in the class.

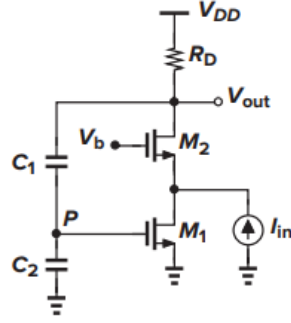


Fig. 3

Refer to Razavi 8.2 and changed question and some parameters.

(a)

VCF

(b)

$$\text{loop gain} = g_{m1} R_D \frac{C_1}{C_1 + C_2}$$

(c)

$$R_{in, closed} = \frac{1}{g_{m2}} \frac{1}{1 + g_{m1} R_D \frac{C_1}{C_1 + C_2}} = 50 \Rightarrow I = 1.946 \text{ mA} \quad (3.1)$$

$$g_{m2} = \sqrt{2 \times I \mu_n C_{ox} \left(\frac{W}{L} \right)_2} = \sqrt{2} g_{m1}, g_{m1} = \sqrt{2 \times I \mu_n C_{ox} \left(\frac{W}{L} \right)_1} = 0.1 \sqrt{I}$$

- 8.4. For the transconductance amplifier circuit in Fig. 4, assume M_1 operates in saturation, $\mu_n C_{OX} = 50 \times 10^{-6} \text{ A/V}^2$, $V_{TH} = 0.7\text{V}$, and the amplifier's input resistance $R_{in} = \infty$. Please answer the following questions:
- Connect V_{in} and the feedback signal V_{FB} to the right terminal of amplifier to form negative feedback, and explain the feedback type. (Choose one from VVF, CVF, VCF and CCF)
 - Assuming that $V_{in,DC} = 1\text{V}$ and the amplifier has low-frequency gain of $A_0 = \infty$, please calculate I_D , g_m and r_o of M_1 . Neglect the body effect.
 - If $A_0 = 50$, calculate values of the open-loop gain, loop gain, and closed-loop gain at low frequency. (Please make sure that unit of your answer is right and use the small-signal parameters you obtained in problem (b) to calculate.)
 - Calculate the open-loop and closed loop output-resistance of the circuit, respectively.

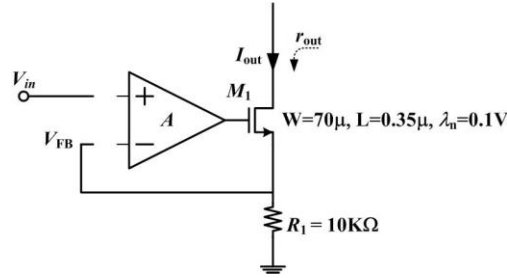


Fig. 4

- (1) V_{in} 接正端, V_{FB} 接负端; CVF
- (2) 运放增益无穷大, 所以运放虚短, $V_{R1} = V_{in} = 1\text{V}$

$$I_D = V_{R1}/R_1 = 100\mu\text{A};$$

$$g_{m1} = \sqrt{2I_D\mu_p C_{ox} \left(\frac{W}{L}\right)} = \sqrt{2 \times 100 \times 50 \times 70 / 0.35} = 1.4\text{mS}$$

$$r_o = \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 100\mu} = 100\text{K}$$

$$(3) G_{open} = i_{out} / v_{in} = \frac{A \times g_m}{1 + g_m \times R_S} = 4.67\text{mS}$$

$$G_{loop} = A \times \frac{g_m R_S}{1 + g_m R_S} = 46.7$$

$$G_{CL} = \frac{G_{open}}{1 + G_{loop}} = 97.9\mu\text{S}$$

- (4) 电流电压反馈

$$r_{out,open} = g_m r_o R_S + r_o + R_S = 1.61\text{M}$$

$$r_{out,close} = r_{out,open} (1 + G_{loop}) = 76.96\text{M}$$